CDA 106
Numerical Methods for Data Science

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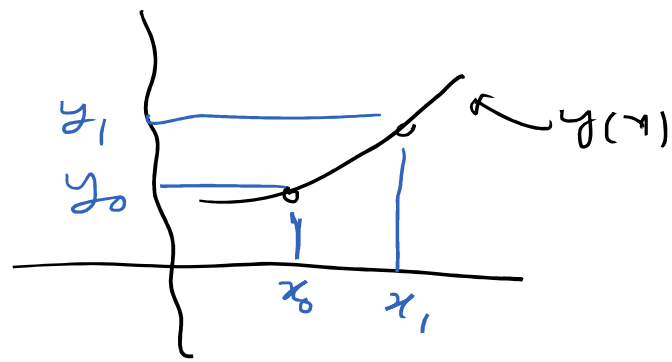
$$\xrightarrow{\qquad\qquad} \qquad\qquad \xleftarrow{\qquad\qquad}$$

$$\begin{array}{cccc} x_0 & x_1 & \cdots & x_n \\ f(x_0) & f(x_1) & \cdots & f(x_n) \end{array}$$

- <u>Forward</u> / <u>Backward</u>.

Therefore, for a given data, we learnt how to find the values of the derivative.

Suppose, $y = y(x)$



Differential eqn

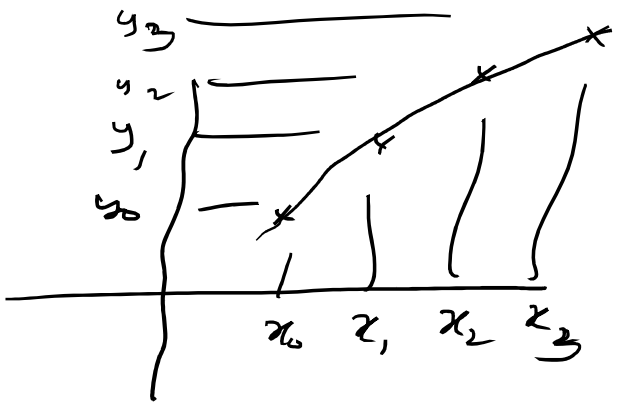
$$\frac{dy}{dx} = f(x, y)$$

In many cases, for example, for $f(x,y) = x^2 + y^2$, above differential eqn cannot be solved (By being solved, I mean to find $y = y(x)$ ~~and~~ as a function of elementary functions)

i.e. $$\boxed{\frac{dy}{dx} = x^2 + y^2}$$ $y = y(x)$??

Sol. For some cases, eqn is ~~not~~ solvable. For example $y' = y$. In this case $y = Ae^x$.

In many cases, however, it may be useful to get the values of y corresponding to the values of x, and not every value $y = y(x)$.



Numerical Sol of Diff-eq^n

$$\{y_i\}_{i \geq 0} = \{y_0, y_1, y_2, \ldots\}$$

To get the numerical sol of differential eq^n, we need to write $\frac{dy}{dx}$ in discrete form.

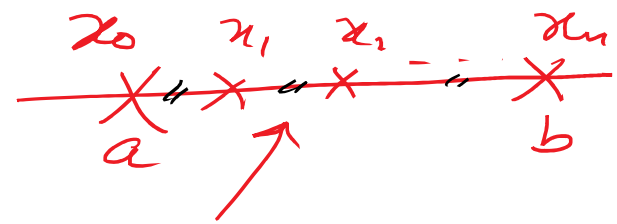
$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{y(x+\Delta x) - y(x)}{\Delta x}$$ By Def^n

Approximations of Derivatives

$$\frac{dy}{dx} \simeq \frac{y(x+\Delta x) - y(x)}{\Delta x}$$

(Expect that Δx is very small)

$$x_i = x_0 + ih = a + ih \qquad 0 \le i \le n$$



$[a, b] \rightarrow n$ parts

$$\frac{b-a}{n} = h$$

First: Forward difference

$$y'(x) = \frac{dy}{dx} \simeq \frac{y(x+h) - y(x)}{h}$$

at any x_i, $y'(x_i) = \dfrac{y(x_i+h) - y(x_i)}{h}$

$$\boxed{y'(x_i) = \frac{y(x_{i+1}) - y(x_i)}{h}} \quad \forall i$$

$0 - - n-1$

Backward difference

$$y'(x_i) \simeq \frac{y(x_i) - y(x_i - h)}{h}$$

$$y'(x_i) = \frac{y(x_i) - y(x_{i-1})}{h} \qquad i = 1, 2, \ldots n$$

Optional

$y'_{(x)} = x^2 + y^2$, $\boxed{y(x_0) = y_0}$

Forward

$$\frac{y(x_{i+1}) - y(x_i)}{h} = x_i^2 + y^2(x_i) \quad \forall i$$

Algebraic eqn and are solvable.

$x_i + h = x_{i+1}$

Euler method.

This completes

Numerical Differentiation

&

Numerical Integration

——— x ——— x ———

Matrix (Plural - Matrices)

A matrix is rectangular array (arrangement) of elements

$$A_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & & a_{2n} \\ \vdots & & & & \\ a_{m1} & a_{m2} & a_{m3} & & a_{mn} \end{bmatrix} = [a_{ij}]_{m \times n}$$

row column

$1 \le i \le m$

$1 \le j \le n$

$m = n$ square matrix

a_{23} = element of row 2 & column 3

System of linear algebraic eqns

$$\left.\begin{aligned} 2x + y &= 3 \\ 2x - y &= 1 \end{aligned}\right\} \quad (*)$$

$$\underbrace{\begin{bmatrix} 2 & 1 \\ 2 & -1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} 3 \\ 1 \end{bmatrix}}_{b}$$

$AX = b$ is matrix eqn corresponding to the system of algebraic eqn $(*)$

- Advantage in finding sol^n

- Assume that A is invertible
  - $|A| \neq 0$
  - A^{-1} exists

Therefore $AX = b$ can be written as

$$A^{-1}(AX) = A^{-1}(b)$$

$$\Rightarrow (A^{-1}A)X = A^{-1}b$$

$$\Rightarrow \boxed{X = A^{-1}b}$$

Sol^n.

Let's solve $A = \begin{bmatrix} 2 & 1 \\ 2 & -1 \end{bmatrix}$ $b = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ $X = \begin{bmatrix} x \\ y \end{bmatrix}$

$$A^{-1} = \frac{\text{Adj}\,A}{|A|} = \frac{(\text{Cofactor}\,A)^T}{|A|} = \frac{1}{-4}\begin{bmatrix} -1 & -1 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} 1/4 & 1/4 \\ 1/2 & -1/2 \end{bmatrix}$$

$|A| = -4$

<u>Therefore</u>

$$X = \begin{bmatrix} 1/4 & 1/4 \\ 1/2 & -1/2 \end{bmatrix}\begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\boxed{x = 1,\ y = 1}$$

Cramer's Rule

$$A_{n\times n} X_{n\times 1} = b_{n\times 1} \quad (*) \qquad X = [x_1, x_2, \dots, x_n]^T$$

$D = |A|$

$D_i = |A_i|$ where A_i is matrix A in which ith column is replaced by $b_{n\times 1}$

Then the sol^n of $(*)$ is given as

$$x_i = \frac{D_i}{D} \quad 1 \le i \le n$$

Example

$$\underbrace{\begin{bmatrix} 2 & 1 \\ 2 & -1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 3 \\ 1 \end{bmatrix}}_{b}$$

$D = -4$ already calculated.

$$D_1 = \begin{vmatrix} 3 & 1 \\ 1 & -1 \end{vmatrix} = -4$$

$$D_2 = \begin{vmatrix} 2 & 3 \\ 2 & 1 \end{vmatrix} = -4$$

$$x = \frac{D_1}{D} = 1$$

$$y = \frac{D_2}{D} = 1$$

Note that in both the methods, the calculations or computation becomes ~~very~~ very complicated when the size of matrix A becomes large

i.e. no of variable / no of eqn are → large

Therefore, when we have large number of eqn is large no of variable, these methods may not be efficient (Time consuming & computationally complex)

→ Which are the other methods which can be used?

First one is "<u>Gauss Elimination Method</u>"

Example

$$2x + y = 3 \quad \text{---(i)}$$
$$2x - y = 1 \quad \text{---(ii)}$$

Eliminate x from eq^n (ii).

For this [(ii) — (i)] →

$$2x + y = 3$$
$$-2y = -2$$

$y = 1$ → $2x + 1 = 3 \Rightarrow \boxed{x = 1}$

$$\boxed{x = 1, \; y = 1}$$

The idea here is, forward elimination & Backward substitution.

Consider Example

$$x_1 + x_2 + x_3 = 6$$
$$2x_1 + x_2 + 3x_3 = 13$$
$$3x_1 + 3x_2 + 4x_3 = 20$$

$$x_1 = 3$$
$$x_2 = 1$$
$$x_3 = 2$$

Convert these eqns into matrix form

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 3 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 13 \\ 20 \end{bmatrix}$$

$R_3 - 3R_1$ // $R_2 - 2R_1$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \\ 2 \end{bmatrix}$$

First we do forward elimination

we keep on doing elimination, until A reduces to upper Δr matrix

Then we apply Backward substitution

$x_3 = 2$

Backward Substitution

$-x_2 + x_3 = 1 \Rightarrow \underline{\underline{x_2}} = x_3 - 1 = 2 - 1 = \underline{\underline{1}}$

$x_1 + x_2 + x_3 = 6$

$x_1 = 6 - x_2 - x_3 = 6 - 2 - 1$
$= 3$

$x_1 + x_2 + x_3 = 6$

$x_1 = 6 - x_2 - x_3 = 6 - 5 = 1$

$x_1 = 1$

x_2

$$\boxed{\begin{aligned} x_1 &= 3 \\ x_2 &= 1 \\ x_3 &= 2 \end{aligned}}$$

Augmented Matrix

System $AX = b$ can be replaced by

$[A \mid b] \leftarrow$ Augmented matrix

Example

$$x_1 + x_2 + x_3 = 6$$
$$3x_1 + 3x_2 + 4x_3 = 20$$
$$2x_1 + x_2 + 3x_3 = 13$$

Matrix form

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 3 & 3 & 4 & 20 \\ 2 & 1 & 3 & 13 \end{array}\right] \xrightarrow[R_3 - 2R_1]{R_2 - 3R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 0 & 1 & 2 \\ 0 & -1 & 1 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$\xleftarrow[\text{Substitution}]{\text{Backward}}$$

$$x_1 = 3$$
$$x_2 = 1$$
$$x_3 = 2$$

Let us recall

$U = \begin{bmatrix} \ddots & \\ 0 & \ddots \end{bmatrix} = [u_{ij}]$

upper triangular matrices

$u_{ij} = 0$ for $i > j$

$L = \begin{bmatrix} \ddots & 0 \\ & \ddots \end{bmatrix} = [l_{ij}]$

lower triangular matrices

$l_{ij} = 0$ for $i < j$

$D = \begin{bmatrix} \ddots & 0 \\ 0 & \ddots \end{bmatrix} = [d_{ij}]$

Diagonal matrix

$d_{ij} = 0 \quad i \neq j$